



STRUCTRA
TRAINING

LIVE ONLINE • SEPTEMBER 2026

Revit for Structures

Learn to model with confidence. Create coordinated concrete and steel structures in Autodesk Revit — then turn them into clear, buildable documentation.

CONCRETE TRACK

STEEL TRACK

NEXT INTAKE

Monday, 14 September 2026

Weekday evenings • Live instructor-led Zoom sessions

NO PRIOR REVIT NEEDED

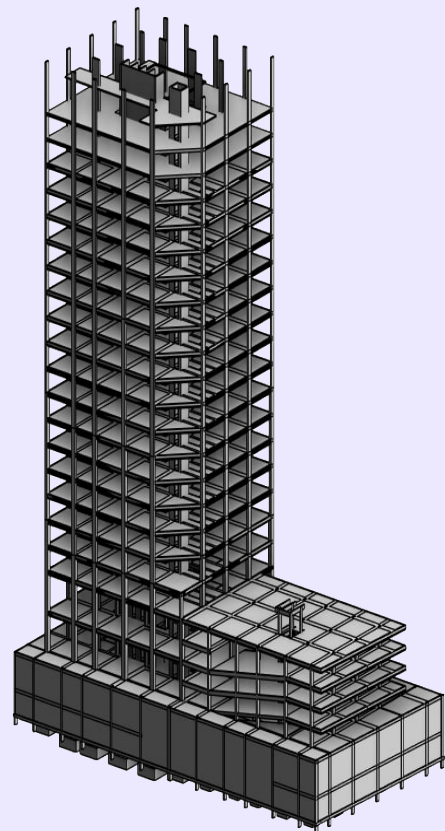
4 weeks
per track

KSH 2,500
per track

100% online
live on Zoom

COURSE OUTLINE / 2026

Model • Detail • Document



Build something real.

Two focused tracks. One practical, portfolio-ready learning experience.



A PRACTICAL PATH INTO STRUCTURAL BIM

Everything you need to build with confidence.

Revit for Structures is a hands-on program for engineering students and early-career professionals who want to build real, coordinated structural models — not just attractive renders. Start with the fundamentals, then progress through modeling, detailing, schedules and construction documentation.

AT A GLANCE



START DATE
Monday, 14 September 2026



FORMAT
100% online
Live sessions via Zoom



DURATION
4 weeks per track
Weekday evenings



FEE
KSH 2,500 per track



DESIGNED FOR
Students + early-career
structural / BIM professionals



WHAT YOU NEED
A Revit-ready laptop
No prior experience required

CHOOSE YOUR TRACK Follow one focused route — or complete both.

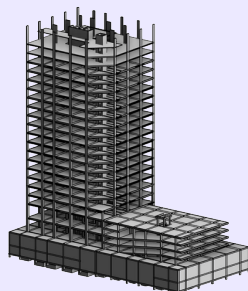
TRACK ONE

Concrete

Modeling • reinforcement • sheets

20 MODULES

● Flexible — choose one or both



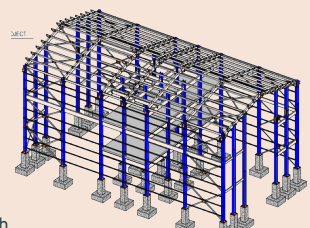
TRACK TWO

Steel

Framing • connections • schedules

29 MODULES

● Flexible — choose one or both





TRACK ONE

Reinforced concrete

MODULES 01–11

Foundations, datum + core structural elements

01 Brief overview of Revit

- Difference with AutoCAD software
- Benefits over AutoCAD software

02 Setting up the software

- Brief introduction of libraries required
- Downloading other extensions from the Autodesk site

03 Beginning with Revit

- Creating a new project
- An overview of the templates
- An overview of palettes, tabs and panels

04 Creating datum elements

- Creating levels
- Creating grids
- Propagating grid extents
- Customizing grid bubble shape and appearance
- Linking CAD files to create grids

05 Overview of creating 3D elements

- Foundations, columns, beams and slabs

06 Creating columns

- Placement of columns
- Loading column families

07 Creating foundations

- Hosting foundations on the columns created
- Creating strip footing
- Using the eccentricity option
- Basement floor slab
- Loading foundation families into your drawings

08 Creating beams and slabs

- Copying beams and slabs from level to level
- Creating a slab
- Hollow pot and waffle slabs

09 Curved beams

- Procedure for modeling curved beams

10 Beam system layout and justification rules

- Using the beam system to model structural beams
- Modeling beams using beam system layout rules
- Modeling beams using beam system justification rules
- Deselecting the beam system

11 Creating staircases and ramps

- Types of staircases
- Creating an architectural ramp and a structural ramp
- Modeling a staircase as a ramp
- Shaft openings

TRACK ONE

Reinforced concrete

MODULES 12–20

Reinforcement, documentation + export

12 Visibility/graphics and filters

- Applying visibility/graphic overrides — method 1
- Applying visibility/graphic overrides — method 2
- Applying visibility/graphic overrides — method 3
- Applying visibility/graphic overrides — method 4

13 Creating roofs and truss placement

- Sloped and gable-ended roofing
- Adding system trusses to your roof
- Editing and creating customized trusses

14 Placing reinforcement

- Creating sections in your drawing
- Foundation reinforcement
- Column reinforcement
- Beam reinforcement
- Slab reinforcement
- Staircase reinforcement
- Ramp reinforcement

15 Annotations

- Adding dimensions
- Adding spot elevations
- Adding spot coordinates
- Adding spot slope

16 Detailing of sections

- Placing breaklines in sections
- Using hatch files to create regions
- Placement of rebar tags on reinforcement
- Adding comments on reinforcement
- Tagging structural elements

17 Schedules and material takeoffs

- Creating bar bending schedules
- Loading shape images into bar bending schedules
- Creating material takeoffs for structural elements
- Sorting and filtering fields in schedules

18 Placing drawings on sheets

- Loading different paper sizes for title blocks
- Using system-loaded title blocks
- Editing and customizing title blocks
- Adjusting text and fields in a title block
- Saving title blocks as a Revit family template
- Duplicate options for drawings
- Placing drawings on sheets

19 Printing sheets to PDF format

- Setting up sheets before printing
- Printing one sheet at a time
- Printing several sheets at a time

20 Converting files into CAD and other formats

- Export options for your drawing
- Transporting your drawing to AutoCAD
- Sending your model for analysis



TRACK TWO

Steel structures

MODULES 01–10

Datum, columns, rafters + framing systems

01 Brief overview of Revit

- Difference with CAD software
- Benefits over CAD software

02 Setting up the software

- Brief introduction of libraries required
- Downloading other extensions from the Autodesk site

03 Beginning with Revit

- Creating a new project
- An overview of the templates
- An overview of palettes, tabs and panels

04 Creating datum elements

- Creating levels
- Creating grids
- Elevation markers
- Propagating grid extents
- Customizing grid bubble shape and appearance (optional)
- Linking CAD files to create grids

05 Modeling steel columns

- Placement of columns
- Loading column families

06 Modeling concrete columns

- Placement of concrete columns
- Loading column families

07 Modeling the foundations

- Hosting foundations on the columns created
- Loading other foundation families into your drawings

08 Modeling the rafters

- Placement of rafters
- Loading steel beam sections
- Attaching columns to the rafters
- Copying rafters from grid to grid

09 Modeling the steel beams

- Loading steel beam sections
- Modeling the floor beams
- Justification rule of the beams
- Copying the floor beams
- Modeling beams at the eaves level

10 Beam system layout and justification rules

- Using the beam system to model structural beams
- Modeling beams using beam system layout rules
- Modeling beams using beam system justification rules
- Deselecting the beam system



TRACK TWO

Steel structures

Composite systems, bracing + connections

MODULES 11–21

11

Composite slab and composite beam

- Modeling the composite beam
- Modeling the composite slab
- Cantilevering the composite slab

12

Floor openings

- Using the shaft opening tool

13

Curved beams

- Procedure for modeling curved beams

14

Sloped / slanted beams

- Modeling sloped beams using the offset tool
- Modeling sloped beams using the framing elevation

15

Purlins and side rails

- Modeling purlins using reference planes
- Modeling purlins using the beam system
- Modeling the side rails

16

Bracings

- Vertical bracings
- Horizontal bracing

17

Visibility/graphics and filters

- Applying visibility/graphic overrides — method 1
- Applying visibility/graphic overrides — method 2
- Applying visibility/graphic overrides — method 3
- Applying visibility/graphic overrides — method 4
- Hiding elements

18

Creating sections

- Types of sections
- Creating sections in your drawing

19

Placing reinforcement

- Setting up the reinforcement cover
- Placing reinforcement at the foundation
- Column reinforcement
- Slab reinforcement

20

Splitting elements

- Splitting elements

21

Introduction to connections

- Uploading the connections
- Applying connections to structural elements
- Propagating the connections



TRACK TWO

Steel structures

Connection detailing, schedules + deliverables

MODULES 22–29

22 Connection types

- Main types of connection
- Haunch connection
- Apex connection
- Base plate connection
- Gusset plate bracing connection
- Gusset plate column-base plate connection
- Purlin / side rail connection
- Single-sided end plate (simple connection)
- Single-sided end plate (moment connection)
- Double-sided end plate (simple connection)
- Fin plate (simple connection)
- Clip angle (simple connection)
- Column splice connection
- Creating custom connections

23 Fabrication elements, modifiers and parametric cuts

- Use and location

24 Annotations

- Adding dimensions
- Adding spot elevations
- Adding spot coordinates
- Adding spot slope
- Creating custom bolt and plate tags

25 Detailing of sections

- Placing breaklines in sections
- Using hatch files to create regions
- Placement of rebar tags on reinforcement
- Adding comments on reinforcement
- Tagging structural elements

26 Schedules and material takeoffs

- Creating bar bending schedules
- Loading shape images into bar bending schedules
- Creating material takeoffs for structural elements
- Sorting and filtering fields in schedules

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